■ Data Catalog — Gold Layer

SQL Data Warehouse Project | Business-level data representation for analytics and reporting

The Gold Layer is the business-level data representation, structured to support analytical and reporting use cases. It consists of dimension tables and fact tables for specific business metrics.

1. gold.dim_customers

Purpose: Stores customer details enriched with demographic and geographic data.

Column Name	Data Type	Description
customer_key	INT	Surrogate key uniquely identifying each customer record in the dimension table.
customer_id	INT	Unique numerical identifier assigned to each customer in the source system.
customer_number	NVARCHAR(50)	Alphanumeric customer code used across CRM and ERP systems (e.g., 'AW00011000').
first_name	NVARCHAR(50)	Customer's first name.
last_name	NVARCHAR(50)	Customer's last name.
country	NVARCHAR(50)	Customer's country, standardized from raw ERP location codes (e.g., 'Australia', 'United States').
marital_status	NVARCHAR(50)	Customer's marital status ('Married', 'Single'), standardized from single-character source codes.
gender	NVARCHAR(50)	Customer's gender ('Male', 'Female'); recorded as 'n/a' where not provided by source.
birthdate	DATE	Customer's date of birth, sourced from the ERP system.
create_date	DATE	Date the customer record was first created in the source CRM system.

2. gold.dim_products

Purpose: Provides descriptive attributes for each product, including category and pricing.

Column Name	Data Type	Description
product_key	INT	Surrogate key uniquely identifying each product record in the dimension table.
product_id	INT	Internal product identifier from the source CRM system.
product_number	NVARCHAR(50)	Structured product code combining category and item identifiers (e.g., 'BK-M82S-38').

Column Name	Data Type	Description
product_name	NVARCHAR(50)	Full descriptive product name, typically including model, color, and size (e.g., 'Mountain-100 Silver- 38').
category_id	NVARCHAR(50)	Category code linking the product to its ERP-defined category (e.g., 'BI_MB', 'CO_RF').
category	NVARCHAR(50)	Top-level product grouping (e.g., 'Bikes', 'Components').
subcategory	NVARCHAR(50)	More specific product grouping within a category (e.g., 'Mountain Bikes', 'Road Frames').
maintenance	NVARCHAR(50)	Indicates whether the product requires periodic maintenance ('Yes'/'No').
cost	INT	Standard unit cost of the product, in whole currency units.
product_line	NVARCHAR(50)	Product line classification derived from source line codes (e.g., 'Road', 'Mountain', 'Other Sales').
start_date	DATE	Date the product became active/available for sale.

3. gold.fact_sales

Purpose: Stores transactional sales data at the line-item level for analytical purposes.

Column Name	Data Type	Description
order_number	NVARCHAR(50)	Unique sales order identifier (e.g., 'SO54496'). Multiple rows can share an order number when an order has several line items.
product_key	INT	Foreign key referencing gold.dim_products.product_key.
customer_key	INT	Foreign key referencing gold.dim_customers.customer_key.
order_date	DATE	Date the order was placed by the customer.
shipping_date	DATE	Date the order was shipped, typically a few days after order_date.
due_date	DATE	Date the customer's payment was due.
sales_amount	INT	Total revenue for the line item (quantity x price), in whole currency units.
quantity	INT	Number of units sold for this line item.
price	INT	Unit price of the product at time of sale, in whole currency units.

Entity Relationships

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gold.fact_sales.customer_key → gold.dim_customers.customer_key
gold.fact_sales.product_key → gold.dim_products.product_key
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This forms a star schema, with gold.fact_sales as the fact table and gold.dim_customers / gold.dim_products as surrounding dimension tables — optimized for BI tools and reporting queries.